Quiz Q1. A system with four unknowns has reduced augmented matrix

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 4 \\ 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

Which statement is correct?

A. It has no solution.

B. It has exactly one solution.

C. It has infinitely many solutions.

D. The matrix does not give enough information.

Quiz Q1 answer. The correct answer is C.

Columns 1, 2 and 4 contain pivots, but column 3 does not.

Thus w_3 is free. Every choice of $w_3 \in \mathbb{R}$ determines the three pivot variables, so the system has infinitely many solutions.

Quiz Q2. In Example 1, $\mathbf{c}_3 = \mathbf{c}_1 + \mathbf{c}_2$. Which vector $\mathbf{z}$ guarantees

$$\mathbf{X}(\mathbf{w} + \mathbf{z}) = \mathbf{X}\mathbf{w}$$

for every weight vector $\mathbf{w}$?

A. $\mathbf{z} = (-1, -1, 1, 0)^{\mathsf{T}}$

B. $\mathbf{z} = (1, -1, 0, 0)^{\mathsf{T}}$

C. $\mathbf{z} = (0, -1, 1, 0)^{\mathsf{T}}$

D. $\mathbf{z} = (0, 0, 0, 1)^{\mathsf{T}}$

Quiz Q2 answer. The correct answer is A.

For $\mathbf{z} = (-1, -1, 1, 0)^{\mathsf{T}}$,

$$\mathbf{X}\mathbf{z} = -\mathbf{c}_1 - \mathbf{c}_2 + \mathbf{c}_3 = \mathbf{0}.$$

Thus $\mathbf{z} \in \mathrm{Nul}(\mathbf{X})$ and

$$\mathbf{X}(\mathbf{w} + \mathbf{z}) = \mathbf{X}\mathbf{w} + \mathbf{X}\mathbf{z} = \mathbf{X}\mathbf{w}.$$